

ESE 17 DYNAMIC OF DPS Solved using STATIONARY SOLUTION

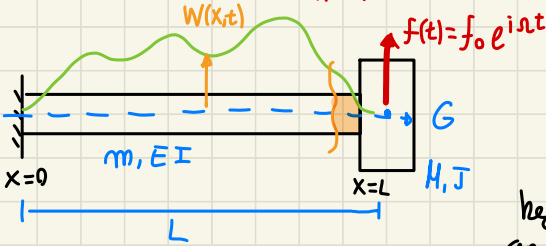
until now we define steady state solution (stationary) for

ARMONIC FUNCTION

free-undamped motion

Now we include also **damping** and **forcing** effect time-dependent

THROUGH MODAL APPROACH



before modal approach application we need the ω_k, ϕ_k of system

{ single uniform mass / material beam L }

+ boundaries • CLAMP (@x=0 CANTILEVER beam)

• LAMPED Rigid body M @x=L (H, J) body

firstly we find nat. freq and modal shapes on a certain Range

implementing the boundaries

$$W(x,t) = [A \cos(\delta x) + B \sin(\delta x) + C \cosh(\delta x) + D \sinh(\delta x)] \cos(\omega t + \varphi)$$

modulate waveform of solution

TRANSVERSAL
MOV of point x

[free undamped motion]

natural
freq of syst.

implement the
matrix and find out
 $\omega, \delta \Rightarrow$ then we

Implement MODAL APPROACH.

find through
problem formulation as linear eq.
4 eq in 4 unk A, B, C, D
 $\Rightarrow$ find δ

Boundaries condition:

@x=0
prevent
ROTATION,
displacement
CLAMP

① $W|_0 = 0$

② $\frac{\partial W}{\partial x}|_0 = 0$

While on the other end @x=L

NO CONSTRAINT, we have a Rigid body attached to the end x=L of beam

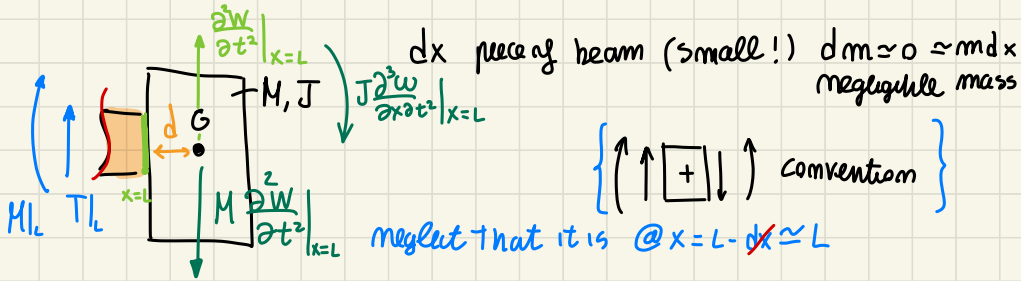
Neither $W, \partial W / \partial x$ @ L are constrained! this end can freely move



NO CONSTRAINT forces @x=L: We can use dyn eq. equation

for last piece of beam attached with H

give us 2 scalar equations to provide ③, ④



if NO RIGID body attached we will have a condition $T_L = M_L = 0$ FREE END of beam

$\frac{\partial^2 W}{\partial x^2} \bigg|_L = \frac{\partial^3 W}{\partial x^3} \bigg|_L = 0$

here INSTEAD we have NON negligible inertia coming from J, M Rigid body

acceleration at G is

$\frac{\partial^2 W}{\partial t^2} \bigg|_{x=L}$ neglecting the small $x=L$, G distance (depending on body size!)

IF we consider also the distance we get complex problem!

so $\ddot{y}_G \approx \frac{\partial^2 W}{\partial t^2} \bigg|_{x=L}$ respect beam

force inertia of M body $F_{in} = -M \frac{\partial^2 W}{\partial t^2} \bigg|_{x=L}$

While the moment of inertia due to J hinge $J \cdot \ddot{\theta}^2$ angular acc of body
body rotation equal to beam one, clamped

$\frac{\partial W}{\partial x} \bigg|_L$ ROTATION displacement

$\frac{\partial^3 W}{\partial x \partial t^2} \bigg|_L$ ANGULAR ACCELERATION pointing ORIENTED as



so moment of inertia forces a opposite

↓
EQUILIBRIUM EQUATION

$\sum M_G = 0$ BODY $T_L \cdot d \approx 0$ moment contribute inertia force & arm

↓ $\left(M_L \right)_{x=L} + J \frac{\partial^3 W}{\partial x \partial t^2} \bigg|_{x=L} = 0$

$EI \frac{\partial^2 W}{\partial x^2} \bigg|_L + J \frac{\partial^3 W}{\partial x \partial t^2} \bigg|_L = 0 \Rightarrow$

$$\textcircled{3} \quad EI \frac{\partial^2 W}{\partial x^2} \Big|_L + J \frac{\partial^3 W}{\partial x \partial t^2} \Big|_L = 0$$

$$\quad \quad \quad \parallel$$

$$\frac{\partial}{\partial x} \left(\frac{\partial^2 W}{\partial t^2} \right) \Big|_L = -\omega^2 \frac{\partial W}{\partial x} \Big|_L$$

$$EI \frac{\partial^2 W}{\partial x^2} \Big|_L - J \omega^2 \frac{\partial W}{\partial x} \Big|_L = 0$$

$\textcircled{4}$ again from D'Alembert on body $M + dx$ piece

$$\sum_i F_y = 0 \rightarrow T|_L - M \frac{\partial^2 W}{\partial t^2} \Big|_L = 0 \quad \text{hence } T = \frac{\partial M}{\partial x} \Big|_L = EI \frac{\partial^3 W}{\partial x^3} \Big|_L$$

$$EI \frac{\partial^3 W}{\partial x^3} \Big|_L - M \left(\frac{\partial^2 W}{\partial t^2} \right) \Big|_L = 0$$

we get

$$A \cos(\delta x) + \dots$$

$$W(x,t) = \alpha(x) \cos(\omega t + \varphi)$$

$$\frac{\partial^2 W}{\partial t^2} = -\omega^2 \alpha(x) \cos(\omega t + \varphi)$$

$$= -\omega^2 W(x,t)$$

$$EI \frac{\partial^3 W}{\partial x^3} \Big|_L + M \omega^2 W \Big|_L = 0$$

NOTICE... on D'Alembert we neglect Mg because our AIM is to find dynamic solution, so weight has just static beam solution small weight effect!

to express the 1,2,3,4 we write $W', W'', W''' \dots$

$$\downarrow \begin{cases} \frac{\partial W}{\partial x} = \gamma [-A \sin(\delta x) + B \cos(\delta x) + C \sinh(\delta x) + D \cosh(\delta x)] \cos(\omega t + \varphi) \\ \frac{\partial^2 W}{\partial x^2} = \gamma^2 [-A \cos(\delta x) - B \sin(\delta x) + C \cosh(\delta x) + D \sinh(\delta x)] \cos(\omega t + \varphi) \\ \frac{\partial^3 W}{\partial x^3} = \gamma^3 [+A \sin(\delta x) - B \cos(\delta x) + C \sinh(\delta x) + D \cosh(\delta x)] \cos(\omega t + \varphi) \end{cases}$$

$\hookrightarrow$

$$\left\{ \begin{array}{l} \textcircled{1} A + C = 0 \\ \textcircled{2} B + D = 0 \\ \textcircled{3} \cancel{\delta^2 EI} [-A \cos(\delta L) - B \sin(\delta L) + C \cosh(\delta L) + D \sinh(\delta L)] + \cancel{\frac{J \omega^2}{\delta^2 EI}} [-A \sin(\delta L) + B \cos(\delta L) + C \sinh(\delta L) + D \cosh(\delta L)] = 0 \\ \textcircled{4} \cancel{\delta^3 EI} [A \sin(\delta L) - B \cos(\delta L) + C \sinh(\delta L) + D \cosh(\delta L)] + \frac{M \omega^2}{\delta^3 EI} [A \cos(\delta L) + B \sin(\delta L) + C \cosh(\delta L) + D \sinh(\delta L)] = 0 \end{array} \right.$$

$$\frac{J \omega^2}{\delta^2 EI} = \frac{J}{m} \delta^2 = C_1$$

$$\frac{M \omega^2}{\delta^3 EI} = \frac{M}{m} \delta = C_2$$

$$[H(\delta L)] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -\cos(\delta L) + C_1 \sin(\delta L) & -\sin(\delta L) - C_1 \cos(\delta L) & \cosh(\delta L) - C_1 \sinh(\delta L) & \sinh(\delta L) - C_1 \cosh(\delta L) \\ \sin(\delta L) + C_2 \cos(\delta L) & -\cos(\delta L) + C_2 \sin(\delta L) & \sinh(\delta L) + C_2 \cosh(\delta L) & \cosh(\delta L) - C_2 \sinh(\delta L) \end{bmatrix}$$

↓
once found the MATRIX [H] of coefficients

↳ **MATLAB** to find f_k and $\phi_k(x)$
nat. freq modal shapes

>> vib-dps-20221122

global variable to share with functions

$$L = 2$$

$$M = m * L / 10 \quad (1/10 \text{ of total beam mass})$$

$$J = M * 0.3^2 \quad (\text{radius of } 0.3 \approx r \text{ of mass})$$

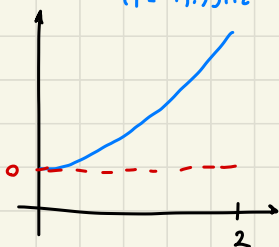
⋮

namefunc = 'cantilever-MJ'

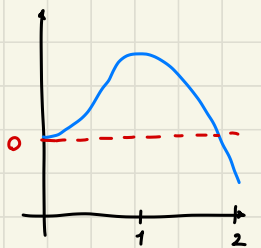
all other is as usual... → {RUN}

$$f_{MAX} = 400 \text{ Hz}$$

$$f_1 = 19.53 \text{ Hz}$$

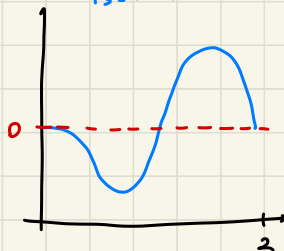


$$f_2 = 118.16 \text{ Hz}$$



same of CANTILEVER
relevant effect of
 $M \downarrow$

$$f_3 = 295.2 \text{ Hz}$$



Increasing mass effect!
with high f , deformation
near $x = L$

similar to first
 $\phi(x)$ of CANTILEVER all
section moving in
phase up/down
 M not so large... affects
small the beam

$\rightarrow$ add $M \uparrow$ without
MORE STIFFNESS $\omega \downarrow$

$\rightarrow$ NEXT PART: consider MODAL APPROACH to define FRF for displacement

of $W(t)|_L$ displacem @ $x=L \rightarrow$ Co-located FRF
(of displacement where we apply the force)

$$W(x,t) = \underline{\phi}(x)^T \underline{q}(t)$$

FRF on a range $(0, 400 \text{ Hz}) \Rightarrow$ on $\underline{\phi}(x)$ we include modal shape of ϕ_1, ϕ_2, ϕ_3

$$\underline{\phi} = \begin{bmatrix} \phi_1(x) \\ \phi_2(x) \\ \phi_3(x) \end{bmatrix} \quad \underline{q}(t) = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

scalar modal cohort of
our W approx, respect modal
approach when $N=3$

considering on theory modal approach of PIN-PIN beam found on LESSON...

$$T = \frac{1}{2} \int_0^L \left(\frac{\partial W}{\partial t} \right)^T m \left(\frac{\partial W}{\partial t} \right) dx$$

$$+ \frac{1}{2} M \frac{\partial W}{\partial t} \bigg|_0 \frac{\partial W}{\partial t} \bigg|_L + \frac{1}{2} J \frac{\partial^2 W}{\partial x \partial t} \bigg|_0 \frac{\partial^2 W}{\partial x \partial t} \bigg|_L$$

on this case
for our EXAMPLE
 $\leftarrow$ represent kinetic energy of beam only
+ we need add mass M due to
kinetic energy ROTOTRANSLATION

kinetic energy

and from T expression because $W = \underline{\phi}^T(x) \underline{q}(t)$

$$\frac{\partial W}{\partial t} = \underline{\phi}^T(x) \dot{\underline{q}}(t) \quad \frac{\partial^2 W}{\partial x \partial t} = \underline{\phi}'^T(x) \dot{\underline{q}}(t)$$

$$\Downarrow$$

$$T = \frac{1}{2} \dot{\underline{q}}^T \left[\int_0^L \underline{\phi}(x) m \underline{\phi}^T(x) dx \right] \dot{\underline{q}} + \frac{1}{2} \dot{\underline{q}}^T \underline{\phi}|_L \underline{\phi}|_L^T M \dot{\underline{q}} + \frac{1}{2} \dot{\underline{q}}^T \underline{\phi}'|_L J \underline{\phi}'|_L^T \dot{\underline{q}} =$$

$$= \frac{1}{2} \dot{\underline{q}}^T \left[\underbrace{\int_0^L \underline{\phi}(x) m \underline{\phi}^T(x) dx + \underline{\phi}|_L M \underline{\phi}|_L^T + \underline{\phi}'|_L J \underline{\phi}'|_L^T}_{[\bar{m}]} \right] \dot{\underline{q}}$$

$[\bar{m}]$ modal mass matrix

and $\forall$ boundaries
condition, ORTHOGONALITY
such that $[\bar{m}]$
 $\Rightarrow$ diagonal MATRIX

{ we have not only distributed mass but also
lumped inertia of M body
due to numerical errors we can have
 $m_{j \neq k} \neq 0$ when $j \neq k$ NO

$\Downarrow$
We force $m_{j \neq k} = 0$ for $j \neq k$ we want find $[\bar{m}]_{3 \times 3}$

to implement on **MATLAB** script

cause $N=3$

after $\phi(x)$ as solution of problem $\rightarrow$ you get $[\bar{m}]$ from here

equally, for modal STIFFNESS and VIRTUAL WORK

modal stiffness Just bending deformation $\Rightarrow$

$$V = \frac{1}{2} \int_0^L \left(\frac{\partial^2 W}{\partial x^2} \right)^T EI \left(\frac{\partial^2 W}{\partial x^2} \right) dx$$

only respect beam
vibrating waves cancel with pre-existing
equilibrium @ $t=0$

if we have lumped springs
another term $\Rightarrow$ we should
include also, $t!$

here NO SPRINGS! so no more to express

$$\frac{\partial^2 W}{\partial x^2} = (\underline{\phi}''(x))^T \underline{q}(t)$$

$$\Rightarrow V = \frac{1}{2} \underline{q}^T \left(\int_0^L \underline{\phi}''(x) EI \underline{\phi}''(x)^T dx \right) \underline{q}$$

$\hookrightarrow [\bar{k}]$ diag modal stiffness

considering damping $\rightarrow$ from h_i^{EXP} experimental non dim ratio

$$h_i^{EXP} = \frac{\bar{C}_{ii}}{C_{iCR}} \quad i=1,2,3 \Rightarrow \bar{C}_{ii} = h_i C_{iCR} = h_i 2 \bar{m}_{ii} \omega_i$$

" found previously " $f: 2\pi$

Virtual work

for a single force $f(t)$ acting:

$$\delta L = \delta W|_L^T f(t) = \delta W|_L = \underline{\phi}^T(L) \delta q$$

$$= \delta q^T \underbrace{\underline{\phi}(L) f_0}_{\underline{Q}_0} e^{i\omega t}$$

system of 3 eq for 3 decoupled single DOF

$$\begin{cases} \bar{m}_{11} \ddot{q}_1 + \bar{c}_{11} \dot{q}_1 + \bar{k}_{11} q_1 = Q_{10} e^{i\omega t} \\ \vdots \end{cases}$$

steady state solution

and from here

$$W|_L = \underline{\phi}(L)^T \underline{q}(t) =$$

$$= \underline{\phi}(L)^T \underline{q}_0 e^{i\omega t} = W_0 e^{i\omega t}$$

$W_0 \in \mathbb{R}$ scalar quantity

$$q_i(t) = q_{i0} e^{i\omega t}$$

$\Leftarrow$

$$q_{i0} = \frac{Q_{i0}}{-\Omega^2 \bar{m}_{ii} + i\Omega \bar{c}_{ii} + \bar{k}_{ii}}$$

$i=1,2,3$

$$FRF = W_0 / f_0 =$$

$$= \phi_1(L) q_{10} + \phi_2(L) q_{20} + \phi_3(L) q_{30} + \dots$$

$\rightarrow$ IMPLEMENT. on MATLAB

once found the solutions
drawing modal shapes

x : set of discrete position across beam length (0,2)

for those x value on W I compute modal shape

$$W_p = \partial W / \partial x$$

$$W_{pp} = \partial^2 W / \partial x^2$$

we can implement the modal parameters

$$\text{havg} \left(\int_0^L \phi(x) m \phi^T(x) dx + \phi|_L \mu \phi|_L^T + \phi|_L J \phi|_L^T \right) = [\bar{m}]$$

inside a for cycle running over the considered $\omega_k \rightarrow k=1,2,\dots,N$

$$\bar{m}_{ii} \left\{ \begin{array}{l} \text{funct} = m * W(i\text{mode}, :) * W(i\text{mode}, :); \\ \text{modal_mass}(i\text{mode}) = \text{trapz}(x, \text{funct}); \\ \quad \quad \quad \text{trapezoidal numerical integration} \\ \text{modal_mass}(i\text{mode}) = \text{modal_mass}(i\text{mode}) + \dots \text{ other terms of problem} \\ \quad \quad \quad \uparrow \\ \quad \quad \quad \bar{m}_{ii} \end{array} \right.$$

$$\bar{k}_{ii} \left\{ \begin{array}{l} \text{funct} = EI * W_{pp}(i\text{mode}, :) * W_{pp}(i\text{mode}, :); \\ \text{modal_stiff}(i\text{mode}) = \text{trapz}(x, \text{funct}) \end{array} \right.$$

↓ + additional possible terms to complete $\bar{k}_{ii}$ if there are springs

$$\begin{aligned} &+ \mu * W(i\text{mode}, \text{end})^2 \\ &+ J W_p(i\text{mode}, \text{end})^2 \end{aligned}$$

↑
($\frac{\partial W}{\partial t}$)

End position,
so @ $x=L$
end section
dependency on where
it's summed

end
↑ finish of cycle overall mat. freq.

once found $\bar{m}_{ii}, \bar{k}_{ii}$

↓
first we check for the mat freq on new system! found through modal approach

matfreq2 → redefine $f_k \quad k=1,\dots,N$

should be $\text{matfreq} \approx \text{matfreq2}$
↑ similar due to numerical method

then we evaluate $\bar{c}_{ii}$

then you implement FRF according to procedure

$f = [0:0.1:f_{MAX}]$;

$F0 = 1$; unitary

$\text{vect-}\omega = \text{zeros}(1, m_f)$;

$\text{ind_input} = \text{length}(x)$ $\leftarrow$ where act the force $f(x)$? where x ?

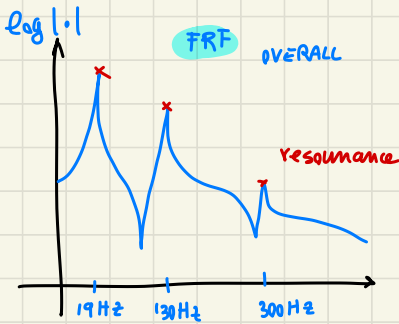
$\text{ind_output} = \text{length}(x)$ $\leftarrow$ where I want find $W(x)$

than on for cycle you compute Q_{oi} (on the script already!)

notice $\bullet /$ $\bullet *$ $\bullet ^2$ $\rightarrow$ for vector operation

than PLOTTING contribute of each freq

and @ end PLOT overall FRF of modes



mode stiffness

$K_{ii} \uparrow \uparrow$ with i so

we need log scale to
see the picks well